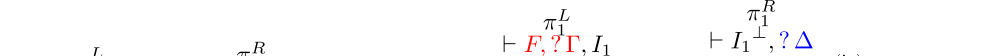
$\vdash !F, ?\Gamma, ?I_1$
π_1^L
$\vdash !I_1^\perp, ?\Delta$
π_1^R
(Cut)
$\vdash !F, ?\Gamma, ?\Delta$
$\longrightarrow_{\mathsf{cut}}$
$\vdash F, ?\Gamma, I_1$
π_1^L
(?d)
$\vdash F, ?\Gamma, I$
$\vdash I_1^\perp, ?\Delta$
π_1^R
(!p)
(Cut)
$\vdash F, ?\Gamma, ?\Delta$
(!p)
$\vdash !F, ?\Gamma, ?\Delta$
$\longrightarrow_{\mathsf{cut}}$ π.